Version: 2.0

TECHNICAL SPECIFICATION

MODEL NO: ES133TT3

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Please contact E Ink or its agent for further information.

☐ Customer's Confirmation

Customer ______________________

Date ______________________

By ______________________

☐ E Ink's Confirmation

Approved By ______________________

Confirmed By ______________________

Prepared By ______________________

Revision History

Rev.	Issued Date	Revised Contents
1.0	2016-07-07	New
2.0	2016-10-11	Revise the drawing of item 4(Mechanical)/13(Packing) Item4: AA dimension tolerance Item 13: Add the BOM list for packing material

TECHNICAL SPECIFICATION

CONTENTS

1. General Description

ES133TT3 is a reflective electrophoretic E Ink technology display module based on active matrix TFT and plastic substrate. The plastic substrate is protected by an outer covering, which is a part of the display. It has 13.3” active area with 2200 x 1650 pixels, the display is capable to display images at 2-16 gray levels (1-4 bits) depending on the display controller and the associated waveform file it used.

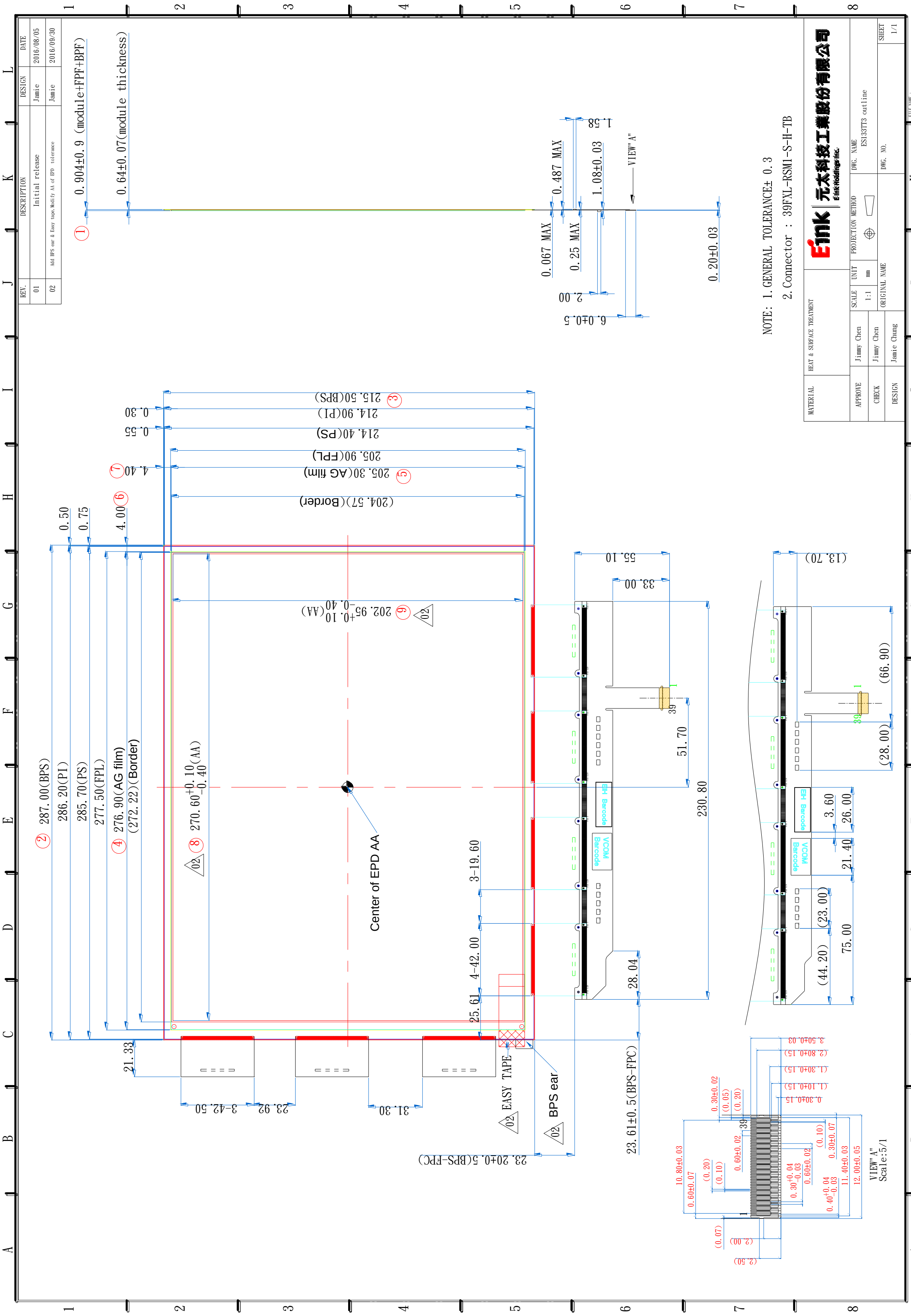
2. Features

- High contrast TFT electrophoretic
- 2200x1650 display
- High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable
- Commercial temperature range
- Antiglare hard-coated front-surface
- Plastic substrate

3. Mechanical Specifications

Parameter	Specifications	Unit	Remark
Screen Size	13.3	Inch	
Display Resolution	2200 (H)×1650(V)	Pixel	
Active Area	270.6(H)× 202.95(V)	mm	
Pixel Pitch	0.123(H)×0.123(V)	mm	
Pixel Configuration	Square		
Outline Dimension	287(W)×215.5(H)×0.64 (D)	mm	
Module Weight	68 ± 7	g	
Number of Grey	16 Grey Level (monochrome)		
Display operating mode	Reflective mode		
Surface treatment	Anti-glare treatment		

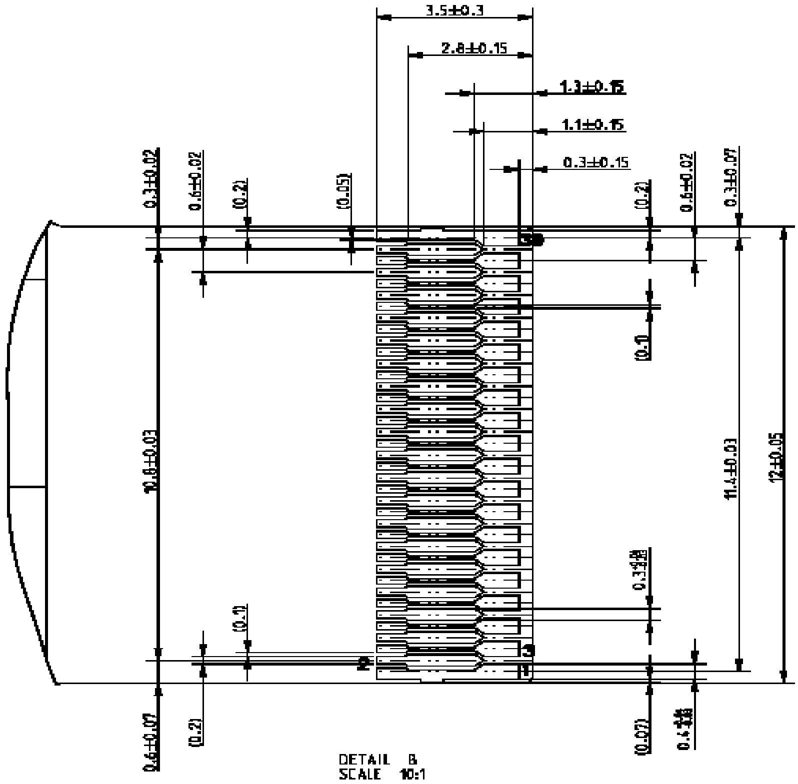
4. Mechanical Drawing of EPD Module



5. Input / Output Interface

5.1 Connector Type

Service	Connector	Type Number	Number of Pins	Mating Connector
Interface	Molex	5025983991	39	Copper foil 0.3mm pitch

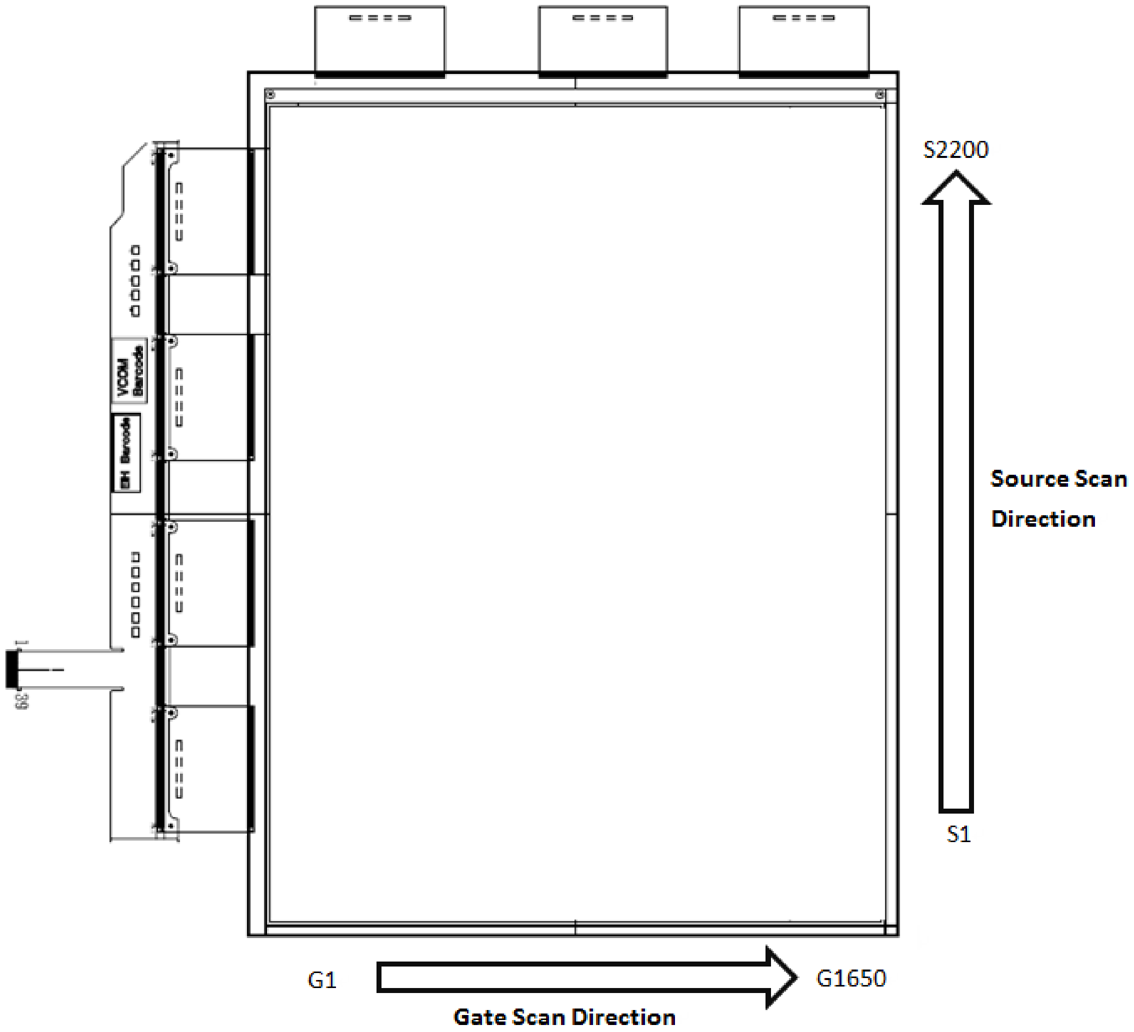


5.1.1 Pin Assignment

Pin #	Signal	Description	Remark
1	VNEG	Negative power supply source driver	
2	VPOS	Positive power supply source driver	
3	VCOM	Common connection	
4	VSS	Ground	
5	VSS	Ground	
6	CKH	Clock source driver	
7	VSS	Ground	
8	LEH	Latch enable source driver	
9	OEH	Output enable source driver	
10	STH	Start pulse source driver	

11	VSS	Ground	
12	D0	Data signal source driver	
13	D1	Data signal source driver	
14	D2	Data signal source driver	
15	D3	Data signal source driver	
16	D4	Data signal source driver	
17	D5	Data signal source driver	
18	D6	Data signal source driver	
19	D7	Data signal source driver	
20	VSS	Ground	
21	D8	Data signal source driver	
22	D9	Data signal source driver	
23	D10	Data signal source driver	
24	D11	Data signal source driver	
25	D12	Data signal source driver	
26	D13	Data signal source driver	
27	D14	Data signal source driver	
28	D15	Data signal source driver	
29	VSS	Ground	
30	MODE1	Output mode selection gate driver	
31	STV	Start pulse gate driver	
32	CKV	Clock gate driver	
33	BORDER	Border connection	
34	VSS	Ground	
35	VDD	Digital power supply drivers	
36	VSS	Ground	
37	VSS	Ground	
38	VEE	Negative power supply gate driver	
39	VGG	Positive power supply gate driver	

5.2 Panel Scan Direction



6. Display Module Electrical Characteristics

6.1 Absolute Maximum Rating

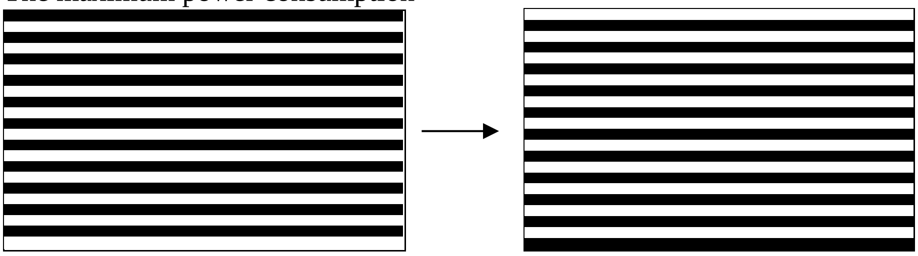
Parameter	Symbol	Rating	Unit	Remark
Logic Supply Voltage	V_{DD}	-0.3 to +7	V	
Positive Supply Voltage	V_{POS}	-0.3 to +18	V	
Negative Supply Voltage	V_{NEG}	+0.3 to -18	V	
Max .Drive Voltage Range	V_{POS} - V_{NEG}	36	V	
Supply Voltage	V_{GH}	-0.3 to +55	V	
Supply Voltage	V_{GL}	-32 to +0.3	V	
Supply Range	V_{GH}-V_{GL}	-0.3 to +55	V	
Operating Temp. Range	T_{OTR}	0 to +50	℃	
Storage Temperature	T_{STG}	-25 to +70	℃	

6.2 Display Module DC Characteristics

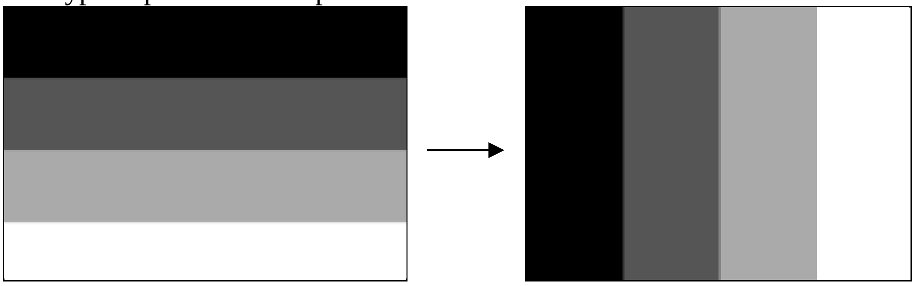
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Signal Ground	V_{SS}			0		V
Logic Voltage Supply	V_{DD}		2.73	3.3	3.6	V
	I_{DD}	V_{DD}=3.3V		3.9	11.5	mA
Gate Negative Supply	V_{GL}		-21	-22	-23	V
	I_{GL}	V_{GL}=-22V		1.9	3.04	mA
Gate Positive supply	V_{GH}		27	28	29	V
	I_{GH}	V_{GH}=28V		2.5	4.84	mA
Source Negative Supply	V_{NEG}		-15.4	-15	-14.6	V
	I_{NEG}	V_{NEG}=-15V		10.7	251	mA
Source Positive Supply	V_{POS}		14.6	15	15.4	V
	I_{POS}	V_{POS}=15V		10.5	248	mA
Border Supply	V_{COM}		-3.5	Adjusted	-0.3	V
Asymmetry Source	V_{ASM}	V_{POS}+V_{NEG}	-800		800	mV
Common Voltage	V_{COM}		-3.5	Adjusted	-0.3	V
	I_{COM}			0.42	0.8	mA
Power Panel	P			440	7730	mW
Standby Power Panel	P_{STBY}				0.4	mW

- The maximum power consumption is measured using 75Hz waveform with following pattern transition: from pattern of repeated 1 consecutive black scan lines followed by 1 consecutive white scan line to that of repeated 1 consecutive white scan lines followed by 1 consecutive black scan lines (Note 7-1).
- The Typical power consumption is measured using 75Hz waveform with following pattern transition: from horizontal 4 grey scale pattern to vertical 4 grey scale patterns (Note 7-2).
- The standby power is the consumed power when the panel controller is in standby mode.
- The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by E Ink.
- V_{COM} is recommended to be set in the range of assigned value ± 0.1V

Note7-1
The maximum power consumption



Note7-2
The typical power consumption

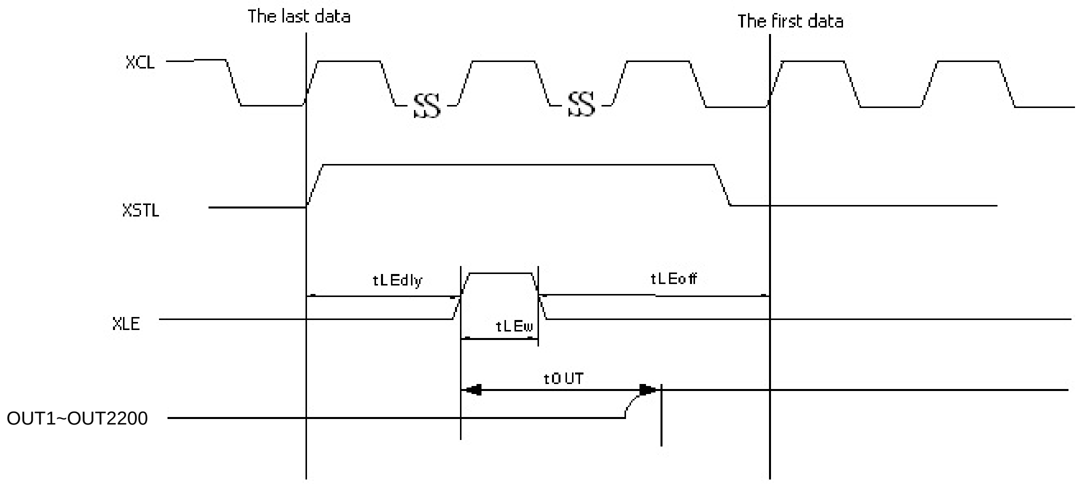


6.3 Display Module AC characteristics

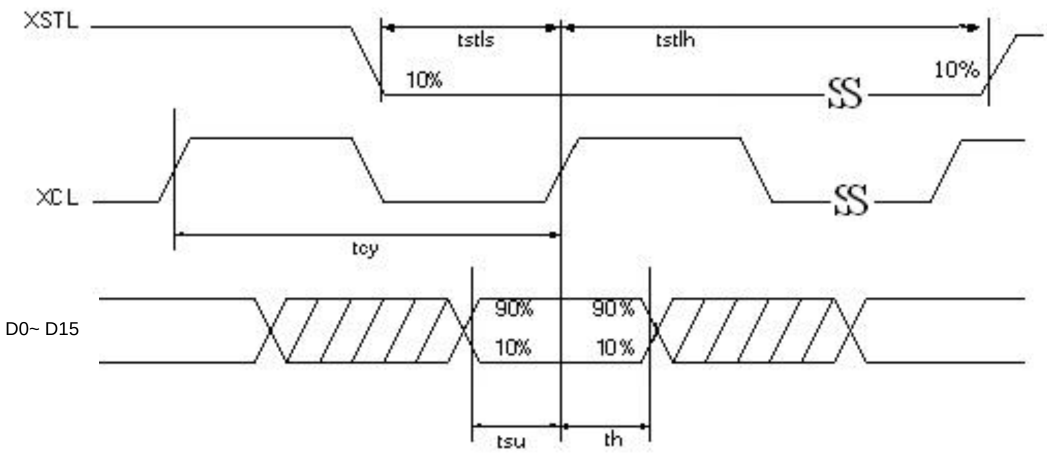
V_{DD}=2.73V to 3.6V, unless otherwise specified.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Clock frequency	fckv	-	-	200	kHz
Minimum "L" clock pulse width	twL	0.5	-	-	us
Minimum "H" clock pulse width	twH	0.5	-	-	us
Clock rise time	trckv	-	-	100	ns
Clock fall time	tfckv	-	-	100	ns
SPV setup time	tSU	100	-	twH-100	ns
SPV hold time	tH	100	-	twH-100	ns
Pulse rise time	trspv	-	-	100	ns
Pulse fall time	tfspv	-	-	100	ns
Clock XCL cycle time	tcy	16.7	-	-	ns
D0 .. D15 setup time	tsu	8	-	-	ns
D0 .. D15 hold time	th	8	-	-	ns
XSTL setup time	tstls	8.35	-	-	ns
XSTL hold time	tstlh	8.35	-	-	ns
XLE on delay time	tLEdly	40	-	-	ns
XLE high-level pulse width (When VDD=2.73V to 3.6V)	tLEw	40	-	-	ns
XLE off delay time	tLEoff	200	-	-	ns
Output setting time to +/- 30mV(C_{load}=200pF)	tout	-	-	12	us
Frame Sync Length (Mode 1)	t1	1			1 line

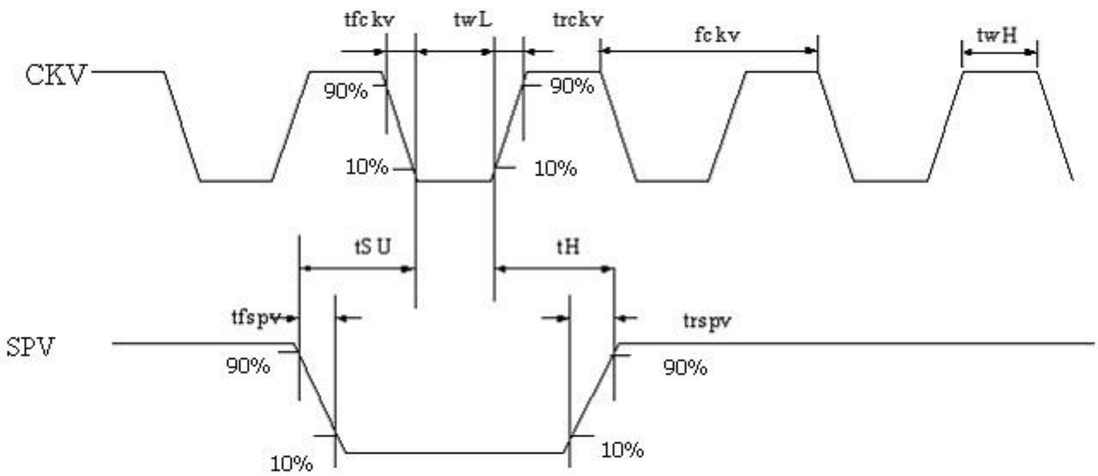
OUTPUT LATCH CONTROL SIGNALS



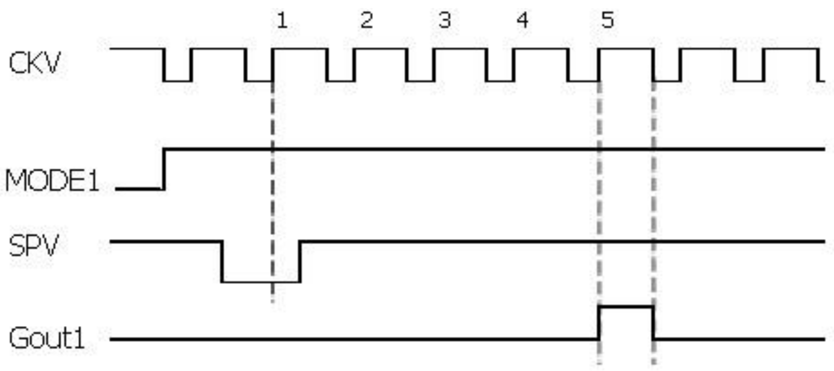
CLOCK & DATA TIMING



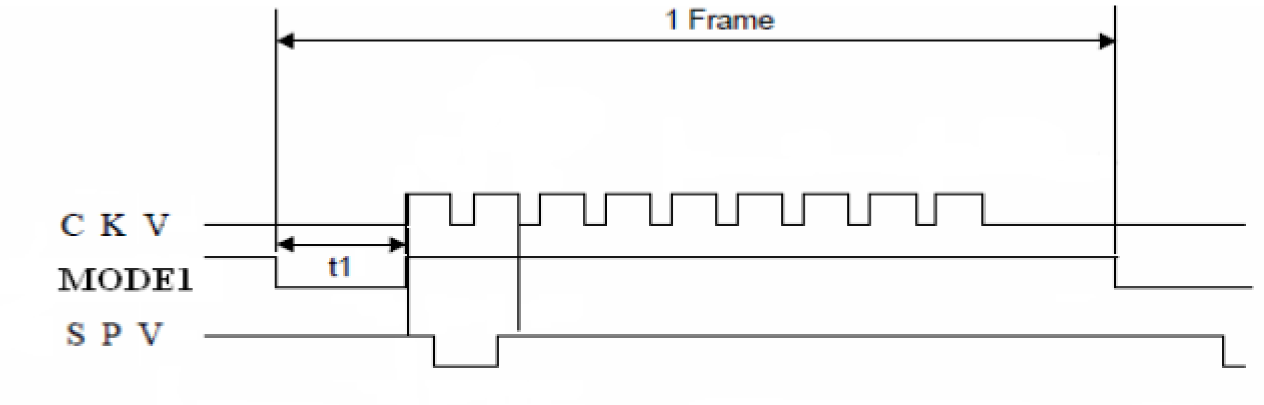
CKV & SPV TIMING



GATE OUTPUT TIMING



FRAME SYNC LENGTH



Note：First gate line on timing

After 5CKV, gate line is on.

6.4 Refresh Rate

The module ES133TT3 is applied at a maximum screen refresh rate of 75Hz.

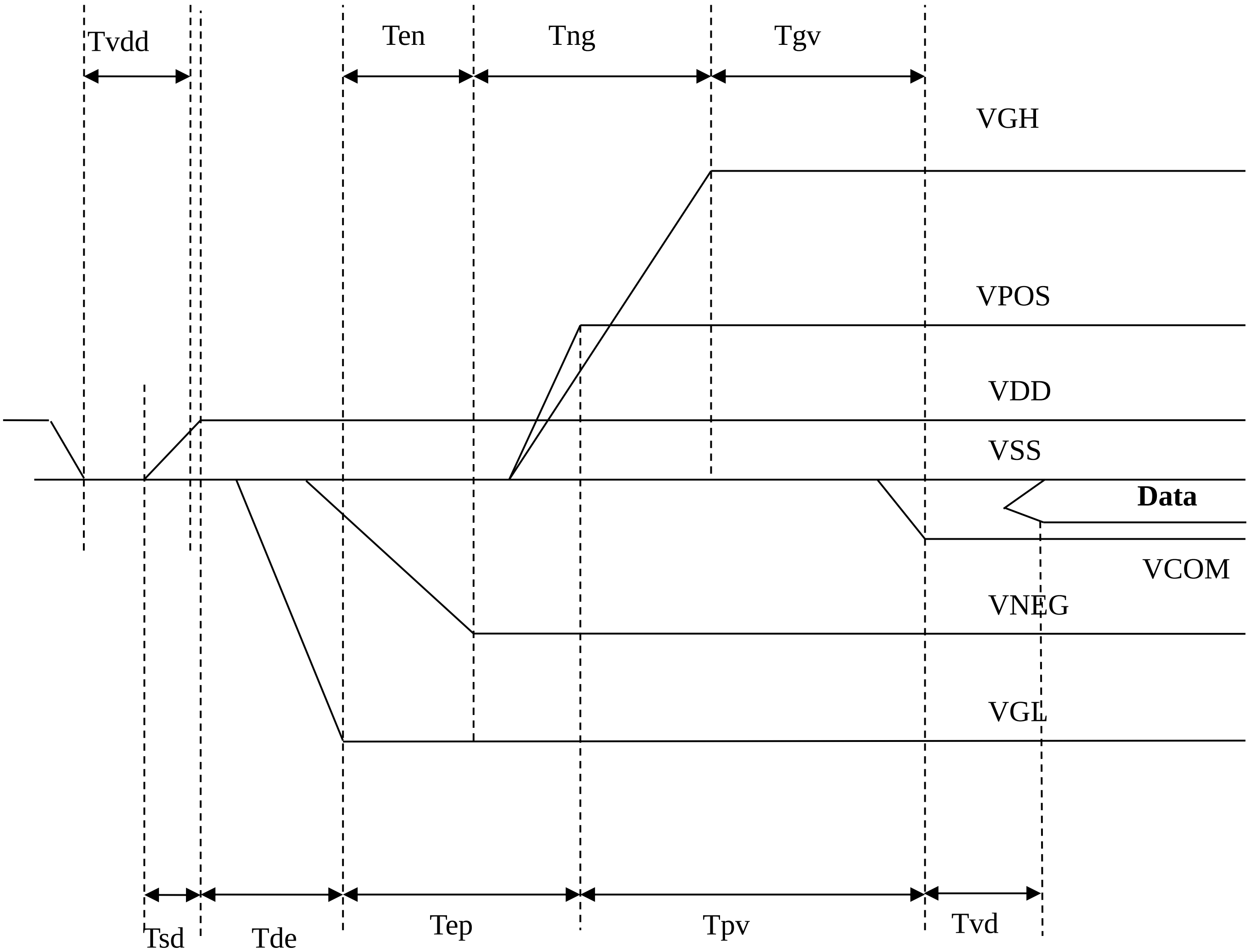
	Min.	**Max.**
Refresh Rate	-	75Hz

7. Power Sequence

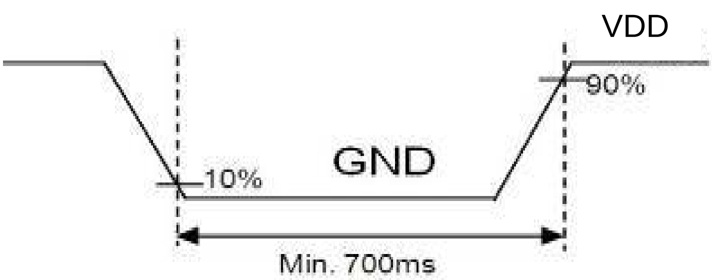
Power rails must be sequenced in the following order：

1. VSS → VDD → VNEG → VPOS (Source driver) → VCOM
2. VSS → VDD → VGL → VGH (Gate driver)

POWER ON

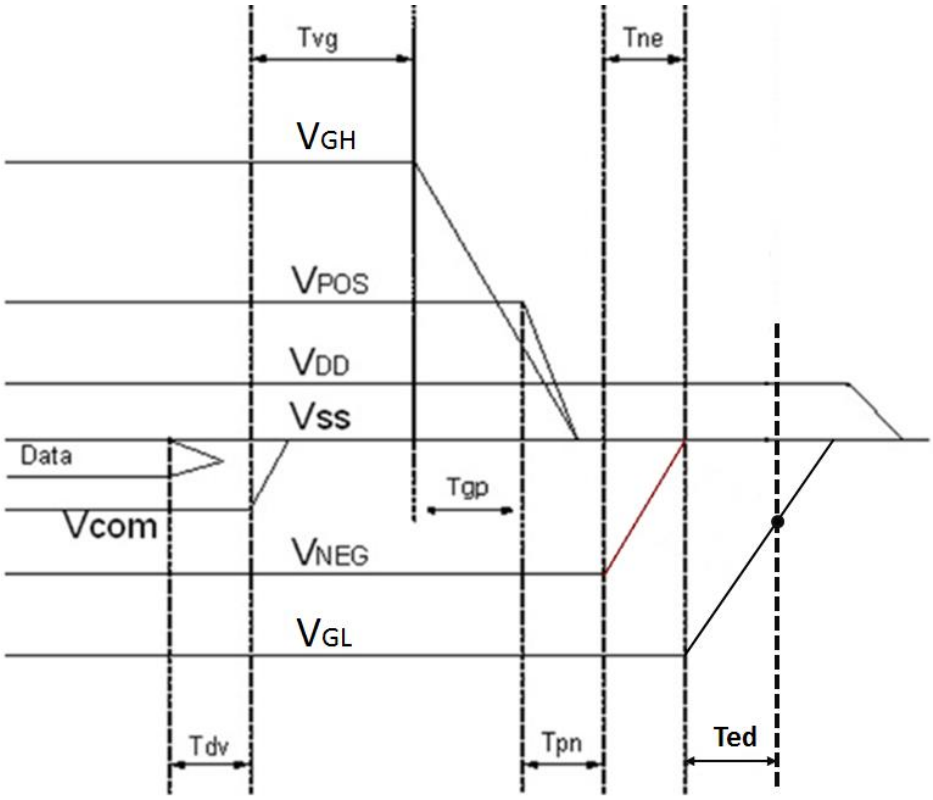


Note: If move from standby mode or power down to power on, VDD voltage must be set GND in the period of 700ms before VDD power on.



	Min.	Max.
Tsd	30us	-
Tde	100us	-
Tep	1000us	-
Tpv	100us	-
Tvd	100us	-
Ten	0us	-
Tng	1000us	-
Tgv	100us	-
Tvdd	700ms	

POWER OFF



	Min.	Max.	Remark
Tdv	100μs	-	-
Tvg	0μs	-	-
Tgp	0μs	-	-
Tpn	0μs	-	-
Tne	0μs	-	-
Ted	0.5s	-	Discharged point @ -7.4 Volt

Note 7-1: Supply voltages decay through pull-down resistors.

Note 7-2: Begin to turn off VGL power after VNEG and VPOS are completely or almost discharged to GND state.

Note 7-3: VGL must remain negative of VCOM during decay period.

8. Optical Characteristics

8.1 Specifications

Measurements are made with that the illumination is under an angle of 45 degrees, the detector is perpendicular unless otherwise specified.

T = 25°C

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit	Note
R	Reflectance	White	30	40	-	%	Note 8-1
Gn	N_{th} Grey Level	-		DS+(WS-DS)× n/(m-1)		L*	-
CR	Contrast Ratio	-	10	12	-		-

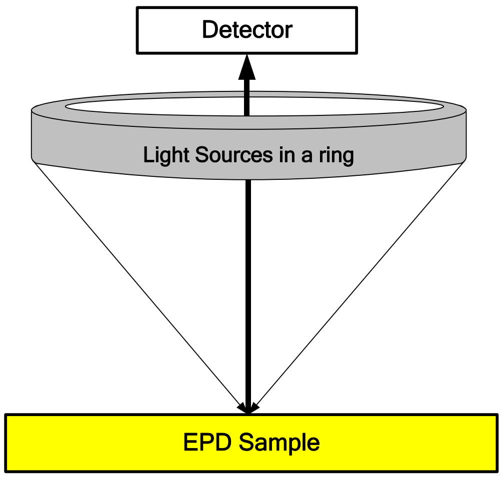
WS: White state , DS: Dark state, Gray state from Dark to White :DS、G1、G2…、Gn…、Gm-2、WS

m:4、8、16 when 2、3、4 bits mode

Note 8-1: Luminance meter: Eye – One Pro Spectrophotometer

8.2 Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (Rl) and the reflectance in a dark area (Rd): CR = Rl / Rd



8.3 Reflection Ratio

The reflection ratio is expressed as:

$R = \text{Reflectance Factor}_{\text{white board}} \times (L_{\text{center}} / L_{\text{white board}})$

L_{center} is the luminance measured at center in a white area (R=G=B=1). $L_{\text{white board}}$ is the luminance of a standard white board.

9. Handling, Safety and Environmental Requirements and Remark

Warning
The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap.

Caution
The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.
Disassembling the display module can cause permanent damage and invalidate the warranty agreements.
IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed.

Mounting Precautions
(1) It`s recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.
(2) It`s recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.
(3) You should adopt radiation structure to satisfy the temperature specification.
(4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.
(5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)
(6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS.
(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.

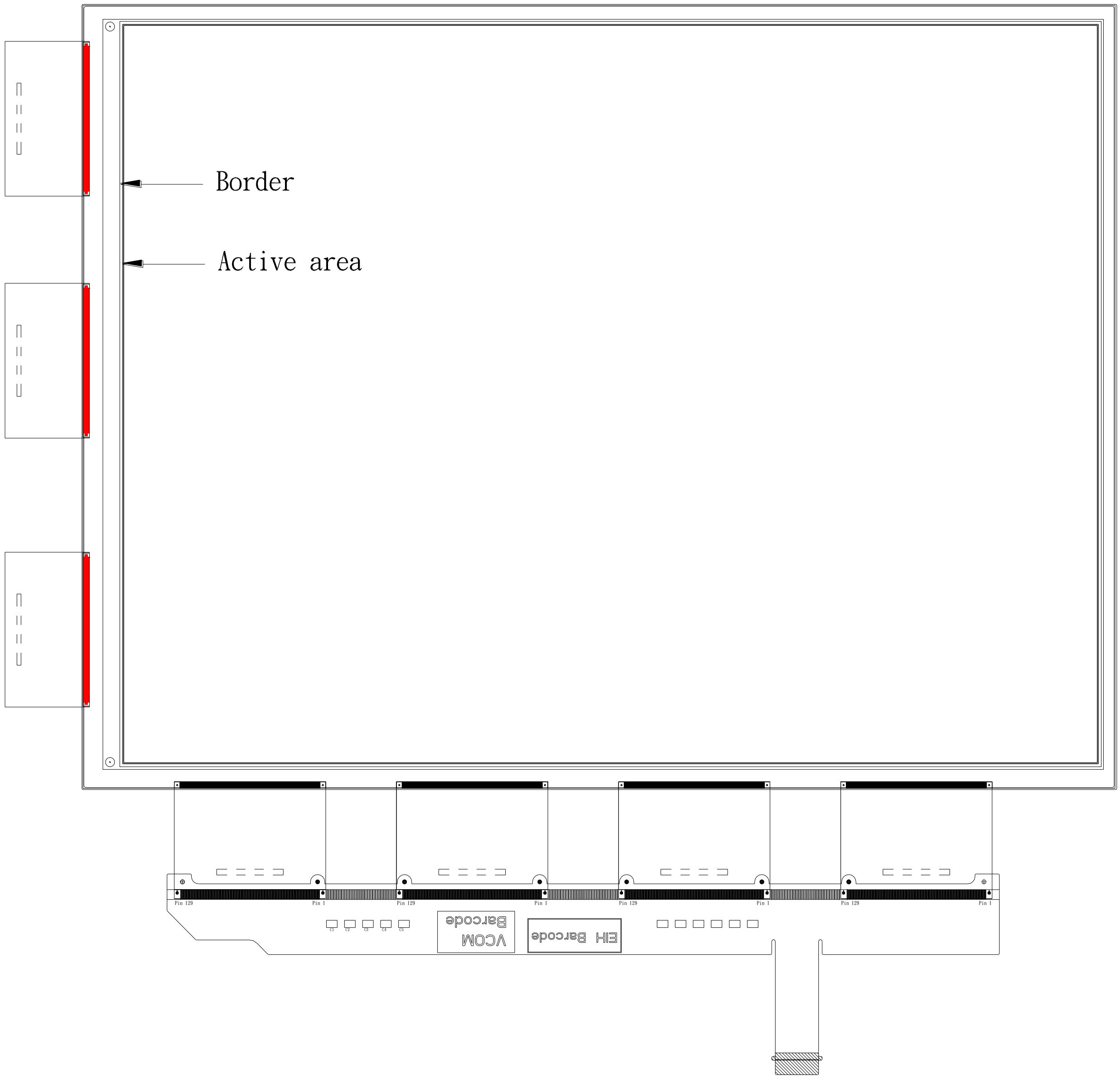
Data sheet status	
Product specification	This data sheet contains preliminary product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	

Remark
All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation.

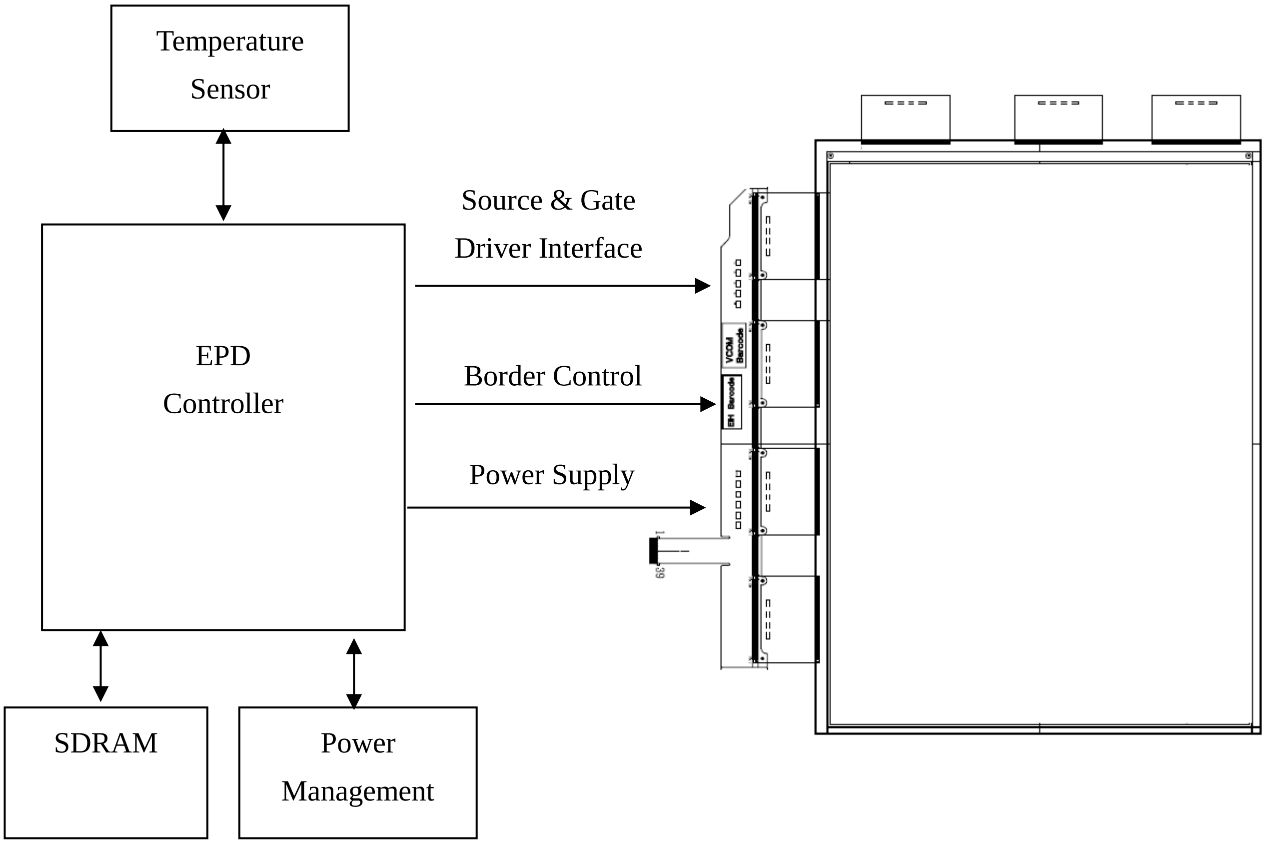
10. Reliability Test

	Test	Condition	Method
1	High Temperature Operation	T=+50C RH=30% for 240hrs	IEC 60 068-2-2Bp
2	Low Temperature Operation	T=0C for 240 hrs	IEC 60 068-2-2Ab
3	High Temperature Storage	T=+70C RH=40% for 120hrs (Test in White Pattern)	IEC 60 068-2-2Bp
4	Low Temperature Storage	T=-25C for 120hrs (Test in White Pattern)	IEC 60 068-2-1Ab
5	High Temperature High Humidity Operation	T=+40C RH=90% for 168hrs	IEC 60 068 2-3CA
6	High Temperature High Humidity Storage	T=+60C RH=80% for 120hrs (Test in White Pattern)	IEC 60 068 2-3CA
7	Temperature Cycle	1 cycle [-25C 30min] -> [+70c 30min] : 100cycles (Test in White Pattern)	IEC 60 068-2-14
8	Solar radiation test	765W/m2 for 168hrs 40C (Test in White Pattern)	IEC 60 068-2-5Sa
9	Electrostatic Efficient (non-operating)	With bezel (Machine model) +/- 250V, 0 ohm , 200Pf	IEC 62179 IEC 62180
10	Package Vibration	1.04G Frequency 10-500Hz Direction: X,Y,Z Duration: 1 hours in each direction	Full package for shipment
11	Package Drop Impact	Drop from height of 122 cm on concrete surface Drop sequence: 1corner, 3edges, 6surfaces One drop for each	Full package for shipment
12	Stylus Tapping	POLYACETAL Pen Top R0.7mm Load:200gf Speed: 30times/min Total: 13,500times	

11. Border Definition

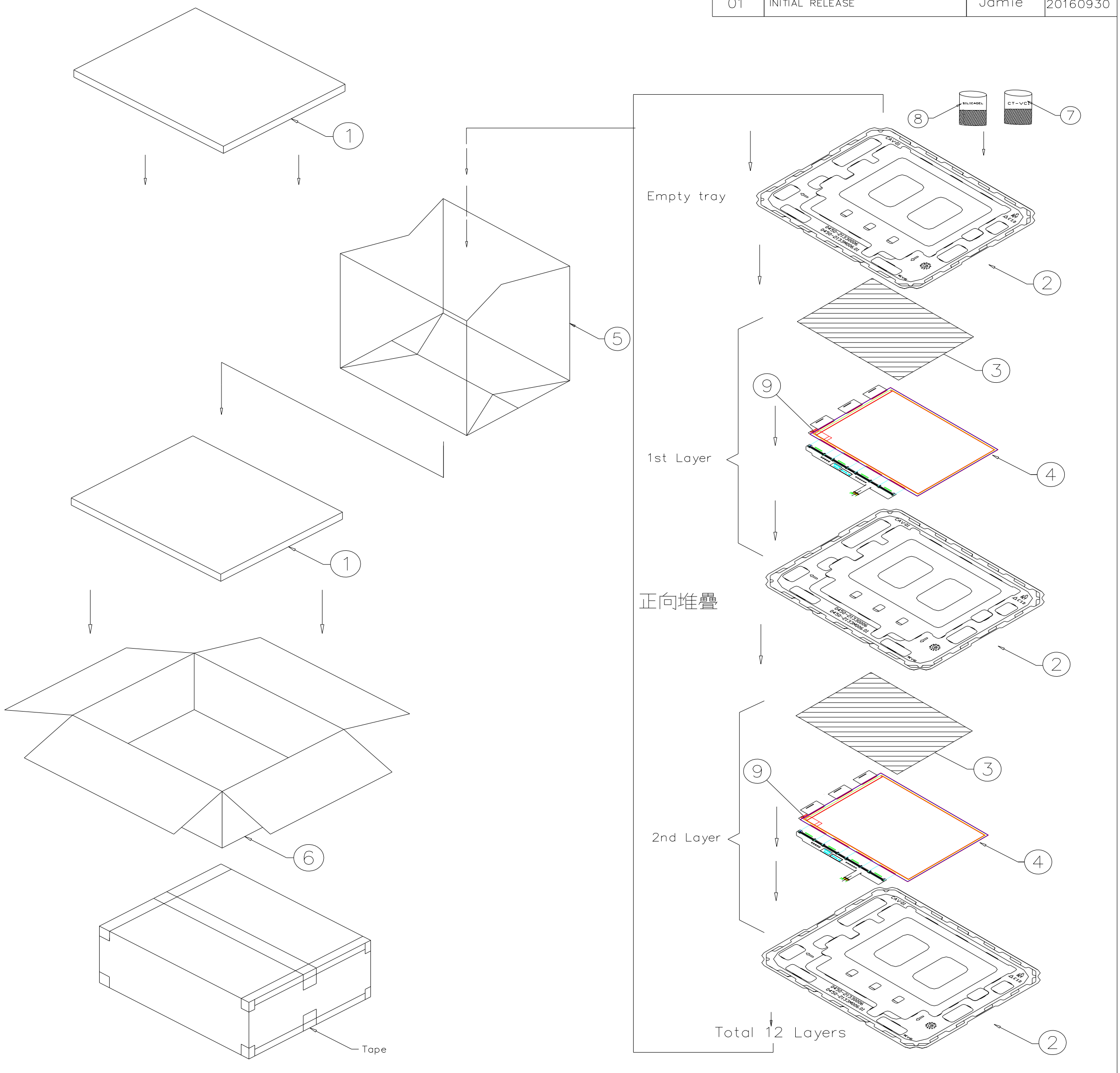


12. Block Diagram



13. Packing

REV	DESCRIPTION	DESIGN	DATE
01	INITIAL RELEASE	Jamie	20160930



NOTE:

1. One layer include: 1 piece of cushion sheet, 1pcs panel & 1 piece of tray.
2. Q'TY: 12 pcs panel/carton.
3. Dimension: 455*375*190mm
4. Weight: 5 KG

ITEM	DESCRIPTION	QTY	REMARK
9	Easy tape	12	
8	30g加厚複合紙活礦乾燥劑73*95mm(禹科JK0030)	2	
7	防鏽劑(保護容積25L)	3	
6	CARTON INTERNAL	1	
5	摺口袋450*380*700mm	1	抗靜電
4	EPD 13.3" Panel	12	
3	EPE CUSHION SHEET	12	抗靜電
2	PET TRAY	13	抗靜電
1	EPE FOAM	2	

MTL.SPEC.	UNSPECIFIED TOL'S ANGLE ROUGHNESS	REMARK	E Ink 元太科技工業股份有限公司 E Ink Holdings Inc.

APPROVE	Jimmy Chen	SCALE 1:1	UNIT mm	SHEET 1 OF 1	DWG.TITLE ES133TT3 PACKING DRAW
CHECK	Jimmy Chen	MTL.NO.			DWG.NO.
DESIGN	Jamie Chung				A4 SIZE